

Keywords: robot manipulation • LLM planning • adaptive control • physical state

Body-Grounded Replanning

Physically Adaptive Robot Manipulation via LLM-Based Internal State Replanning

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arXiv:2609.30024 • September 2026

Body-grounded high-level replanning uses internal physical signals—joint torques and mobility constraints—to trigger LLM-based strategy switching during manipulation, maintaining task success while substantially reducing physical effort under load and asymmetric mobility conditions.

AT A GLANCE

Problem: A manipulation strategy may remain geometrically feasible while becoming physically unsuitable due to increased joint load or limited mobility; current planners have no mechanism to detect or respond to this.

Why it matters: Robots deployed in real environments accumulate physical state changes that degrade performance and increase wear; without adaptation, task success and hardware longevity both suffer.

Method: Body-state events trigger an LLM that interprets joint-level state and execution history to select a physically appropriate alternative strategy from a library, without changing the task objective or low-level controller.

Result: High task success maintained; physical effort reduced; more efficient strategy adaptation than without body state awareness.

Evidence: Reaching task under controlled load and asymmetric mobility constraints, simulation and real robot.

The Big Picture

Robotic manipulation planners reason carefully about the external world: obstacle geometry, object poses, grasp feasibility, and collision avoidance. But they typically treat the robot’s own body as a fixed, ideal actuator whose physical state does not change during execution.

In practice, this assumption fails. A joint under sustained load accumulates fatigue. A component subject to hardware wear constrains the workspace asymmetrically. A strategy that was geometrically and dynamically

feasible at planning time may become physically unsuitable minutes later. The standard response is to run the low-level controller harder—increasing torque outputs to compensate—but this accelerates wear, increases energy consumption, and can ultimately cause failure.

Body-grounded replanning introduces a new level in the robot’s deliberative hierarchy: the ability to recognize when internal physical state has made the current high-level strategy inappropriate, and to select a physically better alternative without changing the task objective or the low-level controller.

Why Existing Approaches Fall Short

Standard task-and-motion planning (TAMP) optimizes over geometry and kinematics. Joint torque, fatigue, and mobility constraints are not modeled at the planning level; they are delegated to the controller.

Adaptive controllers adjust joint-level torques and impedances in response to physical state but do not change high-level strategy (e.g., which direction to approach, which grasp type to use).

LLM-based planners condition on task descriptions and environmental observations—scene graphs, object attributes, occupancy maps—but have no access to joint-level physical signals.

Fault-tolerant planning handles discrete, identifiable hardware failures (a joint locked at zero torque) but not continuous, graded physical state degradation.

Core Mechanism

Body-State Event Detection. Physical state $s_t = (\tau_t, q_t, v_t)$ is monitored continuously. An event triggers when:

- Joint load: $\max_j \tau_{j,t} > \tau_{\text{thresh}}$
- Mobility constraint: reachability $R(q_t) < R_{\text{thresh}}$

LLM Strategy Selector. On event, the LLM receives a prompt encoding:

1. **Joint-level state:** current torques and a natural-language summary of load distribution and mobility limits.
2. **Execution statistics:** success rate and mean completion time of recent strategy attempts.
3. **Execution history:** which strategies were tried and with what outcomes.

The LLM selects a new strategy from the strategy library (e.g., “approach from the side,” “use extended-arm reach,” “shift load to stronger joints”).

Decoupled Architecture. High-level replanning and low-level execution are fully decoupled: replanning changes the strategy but not the task objective or the

controller’s internal parameters.

Technical Formulation

Let $\sigma \in \mathcal{S}$ be the current strategy from library $\mathcal{S}$. Event condition at time t :

$$e_t = \mathbf{1} \left[\max_j \tau_{j,t} > \tau_{\text{thresh}} \vee R(q_t) < R_{\text{thresh}} \right]$$

On $e_t = 1$, the LLM is queried:

$$\sigma_{t+1} = \text{LLM}(\mathcal{P}(s_{t-T:t}, h_{0:t}, \sigma_{\text{hist}}))$$

where $\mathcal{P}$ encodes physical state over window T , execution history h , and prior strategies σ_{hist} .

The optimization objective:

$$\max_{\sigma_{0:\infty}} \Pr[\text{task success}] \text{ s.t. } \sum_t \|\tau_t\|_2 \leq E_{\text{budget}}$$

Learning or Inference Procedure

No model training is required. At execution time:

1. Execute strategy σ with the low-level controller.
2. Monitor s_t at each control step.
3. On $e_t = 1$: construct prompt $\mathcal{P}$, query the LLM.
4. Update $\sigma \leftarrow \sigma_{t+1}$ and resume.
5. Repeat until success or timeout.

The LLM runs at inference time only, with no task-specific fine-tuning.

What the Guarantee Says

The framework does not require training a new model. Its correctness relies on: (i) the strategy library covering alternative strategies that are physically better in the relevant conditions, and (ii) the LLM’s ability to interpret physical state descriptions and select appropriate alternatives. Evaluation shows that both conditions hold for the tested configurations.

Experimental Findings

- **Task:** Reaching under two constraint types—controlled joint load (added deadweight) and asymmetric mobility (reduced joint range)
- **Platforms:** Simulated and real robot arm
- Body-grounded replanning maintains high task success rates where fixed-strategy baselines fail under increasing load
- Cumulative joint torque is substantially reduced by selecting strategies that distribute load across available joints

- Strategy adaptation is more efficient (fewer failed attempts before success) than random strategy switching when mobility is asymmetric
- Performance is robust across multiple off-the-shelf LLMs; stronger models provide modestly better strategy selection

Ablations and Interpretation

Event threshold: Too low τ_{thresh} causes excessive re-planning; too high causes physical overload. Performance is robust within a reasonable range.

History window: More execution history improves strategy selection for mobility constraints; load constraints show less sensitivity.

Strategy library size: A larger library with more alternatives improves success under diverse constraints; the LLM selects appropriately among them.

Architecture decoupling: The clean separation between replanning and execution means the framework can be added to any existing manipulation system without modifying the low-level controller.

Reference

Namiko Saito, Hiroshi Kera. **Body-Grounded Replanning for Physically Adaptive Manipulation**. arXiv:2609.30024, September 2026.
<https://arxiv.org/abs/2609.30024>